

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JnanaSangama”, Belgaum -590014, Karnataka.



MINI PROJECT REPORT

on

“AI POWERED RETINAL DISEASE DIAGNOSIS USING OCT IMAGES”

Submitted by

ATHMICA (1BM22CS058)

BHARATH (1BM22CS068)

BHUVANA(1BM22CS071)

CHRIS DINESH(1BM22CS080)

Under the Guidance of

Prof. Namratha M

Assistant Professor, BMSCE

in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING



B. M. S. COLLEGE OF ENGINEERING

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B. M. S. College of Engineering,
Bull Temple Road, Bangalore 560019
(Affiliated To Visvesvaraya Technological University, Belgaum)
Department of Computer Science and Engineering



CERTIFICATE

This is to certify that the project work entitled “**AI POWERED RETINAL DISEASE DIAGNOSIS USING OCT IMAGES**” carried out by **ATHMICA N (1BM22CS058), BHARATH C (1BM22CS068), BHUVANA M (1BM22CS071) AND CHRIS DINESH(1BM22CS080)** who are bonafide students of **B. M. S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visveswaraiyah Technological University, Belgaum during the year 2023-2024. The project report has been approved as it satisfies the academic requirements in respect of **Mini Project (23CS5PWMIP)** work prescribed for the said degree.

Signature of the Guide
Prof. Namratha M
Assistant Professor
BMSCE, Bengaluru

Signature of the HOD
Dr. Kavitha Sooda
Professor & Head, Dept. of CSE
BMSCE, Bengaluru

External Viva

Name of the Examiner

Signature with date

1. _____

2. _____

B. M. S. COLLEGE OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



DECLARATION

We, ATHMICA N(1BM22CS058), BHARATH C (1BM22CSCS068), BHUVANA M(1BM22CS071), CHRIS D (1B22CS080), students of 5th Semester, B.E, Department of Computer Science and Engineering, B. M. S. College of Engineering, Bangalore, here by declare that, this Project Work-1 entitled "Project Title" has been carried out by us under the guidance of Prof. Namratha M, Assistant Professor, Department of CSE, B. M. S. College of Engineering, Bangalore during the academic semester September 2024- January 2025.

We also declare that to the best of our knowledge and belief, the development reported here is not from part of any other report by any other students.

Signature

ATHMICA
(1BM22CS058)

BHARATH
(1BM18CS068)

BHUVANA
(1BM22CS071)

CHRIS
(1BM22CS080)

1. Introduction

AI-powered retinal disease diagnosis using Optical Coherence Tomography (OCT) images leverages advanced machine learning (ML) and deep learning (DL) techniques to enhance medical imaging analysis. OCT captures high-resolution cross-sectional images of the retina, enabling early detection of retinal diseases such as diabetic retinopathy, age-related macular degeneration (AMD), and glaucoma. AI systems, particularly those based on deep learning models like convolutional neural networks (CNNs), excel at analyzing these images to identify patterns and abnormalities. This technology promises faster and more accurate diagnoses, reducing the burden on ophthalmologists and improving patient outcomes.

1.1 Motivation

Retinal diseases, if undetected early, can lead to irreversible vision loss. Traditional diagnostic methods require skilled professionals and can be time-consuming. By automating the analysis of OCT images with AI, it becomes possible to address the growing demand for timely and accurate diagnoses, especially in regions with limited access to specialized healthcare. The motivation behind this project is to harness AI's potential to enhance diagnostic precision, improve early detection rates, and ultimately contribute to better management of retinal health.

1.2 Scope of the Project

The project focuses on developing an AI-based system capable of analyzing OCT images for detecting and classifying retinal diseases such as diabetic retinopathy, AMD, and glaucoma. It aims to:

1. Automate image analysis to reduce the workload on ophthalmologists.
2. Provide accurate and early detection of retinal conditions.
3. Assist in decision-making by highlighting critical abnormalities in retinal layers.
4. Improve accessibility to high-quality diagnostic tools in underserved areas.

1.3 Problem statement

To develop an AI-based system using machine learning and deep learning techniques that can automatically analyze Optical Coherence Tomography (OCT) images to detect and classify retinal diseases such as diabetic retinopathy, AMD, and glaucoma. The system must accurately identify disease-related patterns in retinal layers, facilitate early diagnosis, and assist ophthalmologists in making quicker and more reliable decisions, ultimately improving patient outcomes.

2. Literature Survey

- **Artificial Intelligence for Early Diagnosis of Retinal Diseases (2024)**
This study employs deep learning models, including CNNs and U-Net architectures, to analyze retinal OCT images. The models achieve high diagnostic performance for diseases like diabetic retinopathy and AMD. The research highlights scalability in resource-limited settings but emphasizes the need for diverse datasets for broader application and improved generalization.
- **Retinal Disease Detection Using Deep Learning Techniques: A Comprehensive Review (2023)**
This paper reviews CNNs and Vision Transformers (ViTs) for detecting diabetic retinopathy, AMD, and glaucoma. While CNNs excel at feature extraction, ViTs offer promise despite higher computational complexity. The paper underscores the importance of data diversity to mitigate overfitting and enhance the reliability of AI models.
- **The Upcoming Role of AI for Retinal and Glaucomatous Diseases (2022)**
This study applies deep learning to fundus photographs, visual fields, and OCT images for diagnosing retinal diseases. It highlights high accuracy in aiding non-specialists but notes performance issues with variable image quality and limited dataset generalization.
- **Artificial Intelligence-Based System for Retinal Disease Diagnosis (2024)**
The research develops a CNN-based framework using datasets like EyePACS and Messidor. It demonstrates high accuracy in disease classification while addressing challenges in generalization due to biases in training data. This emphasizes the need for high-quality datasets for consistent results.
- **Artificial Intelligence in Retinal Disease: Clinical Application, Challenges, and Future Directions (2023)**
This paper discusses the clinical deployment of AI for retinal diseases, focusing on CNNs and RNNs. Despite promising results in sensitivity and specificity, the research highlights the "black box" problem and data quality concerns as barriers to clinician trust and adoption.
- **A Comparative Investigation of Transfer Learning Frameworks Using OCT Images for Retinal Disease Identification (2024)**
This study compares MobileNetV2, EfficientNet, InceptionNetV3, and DenseNet using transfer learning on OCT datasets. MobileNetV2 achieves 100% accuracy, showcasing the potential of transfer learning for precise automation. However, dataset diversity limitations remain a key concern.
- **Retinal Disease Classification from OCT Images Using Deep Learning Algorithms (2023)**
This research employs CNN architectures like VGG16 and DenseNet121 for multi-class disease classification from OCT images. The proposed models achieve high sensitivity and specificity but face challenges in balancing datasets and incorporating data augmentation.
- **Performance Evaluation of Retinal OCT Fluid Segmentation, Detection, and Generalization (2024)**
This paper evaluates models like nnUNet and SAMedOCT for fluid segmentation in OCT

images. While achieving strong performance metrics, the study identifies difficulties in generalizing across devices due to variations in image quality.

- **Artificial Intelligence for Screening of Multiple Retinal and Optic Nerve Diseases (2022)**

A deep learning-based system, RAIDS, is introduced for detecting ten retinal diseases using fundus images. The model outperforms ophthalmologists in accuracy and efficiency but requires high-quality images and extensive datasets for reliable deployment.

- **Artificial Intelligence Techniques in Inherited Retinal Diseases: A Review (2024)**

The study reviews AI techniques for rare inherited retinal diseases like Stargardt disease and Retinitis Pigmentosa. CNNs and U-Net excel in prediction and segmentation but depend heavily on large, well-annotated datasets for reliable performance.

- **A Deep Learning-Based Framework for Retinal Disease Classification (2023)**

Using transfer learning with VGG-19, this study achieves 99.17% accuracy in OCT-based disease classification. It highlights the advantages of transfer learning in reducing training time but notes the risks of overfitting with smaller datasets.

- **Accuracy and Feasibility of AI-Assisted OCT in Community Screening (2022)**

This research demonstrates the effectiveness of AI-assisted OCT systems in detecting multiple retinal disorders during community screenings. High accuracy and specificity are achieved, but challenges arise in sensitivity for less common conditions.

- **A Novel Approach for Diabetic Retinopathy Screening Using Asymmetric Deep Learning Features (2023)**

The study uses U-Net for optic disc segmentation and CNN-SVM for lesion classification in diabetic retinopathy. While achieving high sensitivity and specificity, the model's scalability and generalization are limited by dataset uniformity.

- **Detection of Diabetic Retinopathy Using Fusion of Textural and Ridgelet Features (2024)**

This paper combines textural features with Ridgelet transforms and uses SMO classifiers for DR detection. Despite achieving high accuracy, the method's computational complexity limits its application in large-scale scenarios.

- **Automated Detection of Severe Diabetic Retinopathy Using Deep Learning (2024)**

Employing Inception V3, this study achieves high sensitivity and specificity for severe DR detection. However, distinguishing moderate cases and handling certain lesion types, like IRMAs, remain challenging for accurate severity grading.

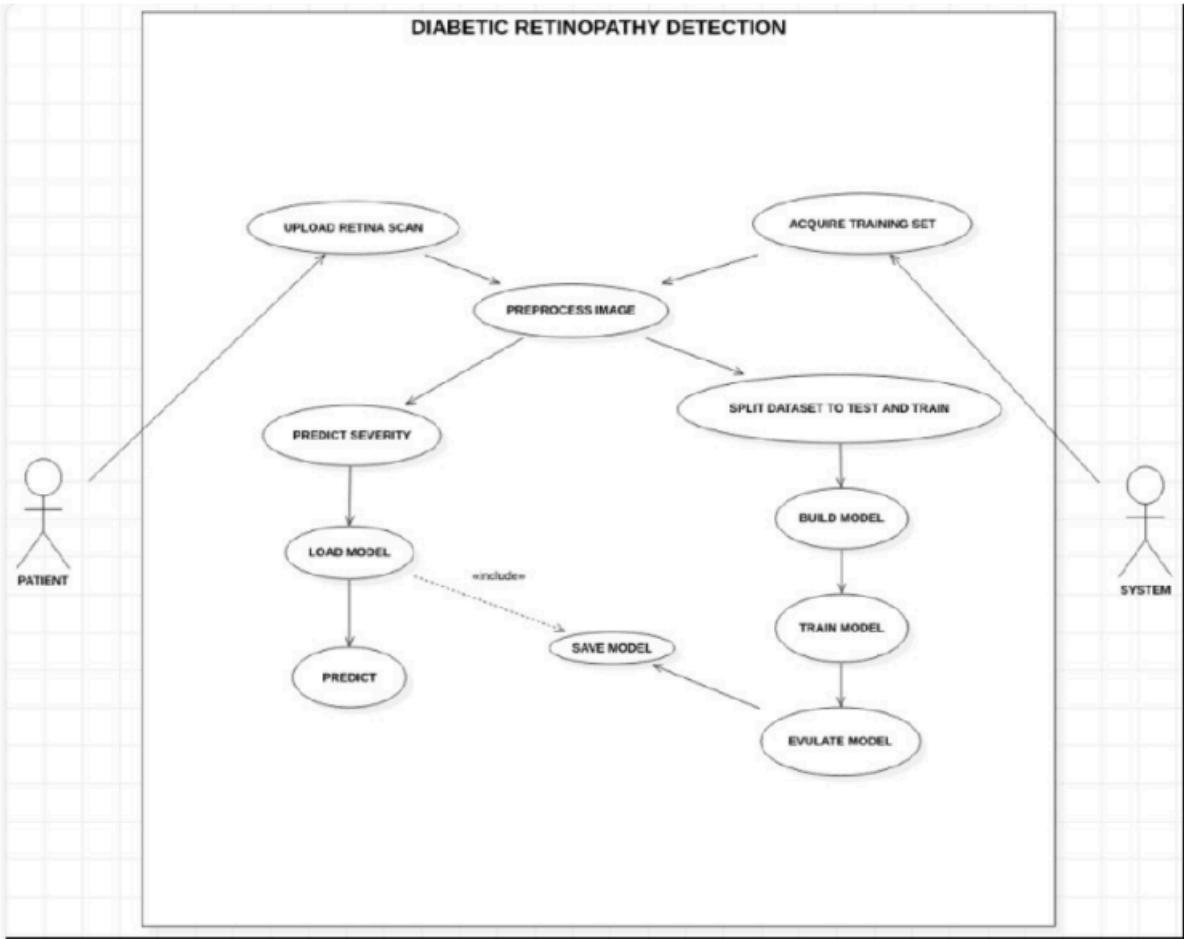
3. Design

3.1 High Level Design

The high-level design provides a top-down view of the system, highlighting its key components and their interactions. The system is designed to perform **diabetic retinopathy classification** by processing input images through a pre-trained **EfficientNetB0** model and a custom classification head. The system workflow includes data ingestion, preprocessing, model training, fine-tuning, and evaluation.

1. **Input Data:**
 - **Images:** Retinal fundus images stored in train_images.
 - **Labels:** Classification labels (0–4) stored in train.csv.
2. **Data Preprocessing:**
 - Prepares the dataset for training and validation.
 - **Data Augmentation:** Random transformations (rotation, zoom, flips, etc.) are applied to improve generalization.
 - **Normalization:** Pixel values are scaled to [0, 1] using rescale=1./255.
3. **EfficientNetB0 Backbone:**
 - Acts as the feature extractor with pre-trained ImageNet weights.
 - Initially frozen to preserve learned features.
4. **Custom Head:**
 - **Global Average Pooling (GAP):** Reduces spatial dimensions.
 - **Dropout:** Prevents overfitting.
 - **Dense Layer:** Outputs class probabilities for 5 classes using softmax activation.
5. **Training:**
 - **Phase 1:** Train only the custom head with the backbone frozen.
 - **Phase 2:** Fine-tune the entire model by unfreezing the backbone layers.
6. **Evaluation:**
 - Predictions are generated for the validation dataset.
 - **Quadratic Weighted Kappa Score** is calculated to evaluate performance.

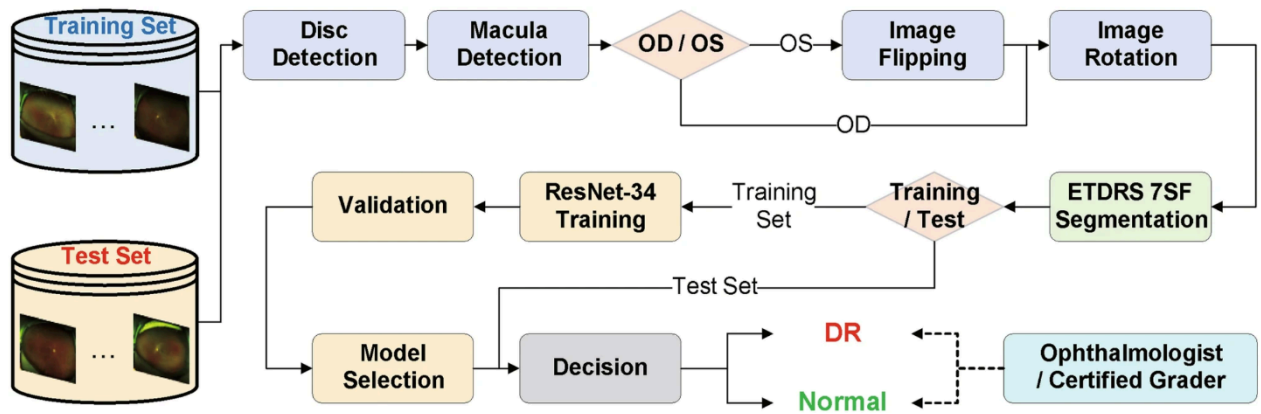
Use case Diagram:



3.2 Detailed Design

The detailed design delves deeper into each component, showing how the system processes the data, trains the model, and evaluates its performance.

1. **Data Loading:**
 - Reads train.csv to extract image filenames and corresponding class labels.
 - Images are loaded from the train_images directory.
2. **Data Preprocessing:**
 - **Augmentation:**
 - Random rotations, zooms, shifts, and flips to improve robustness.
 - **Normalization:**
 - Rescales image pixel values to [0, 1] using Image Data Generator.
3. **Model Architecture:**
 - **EfficientNetB0:**
 - Acts as the feature extractor.
 - Input: (224, 224, 3) pre-processed RGB images.
 - **Custom Head:**
 - **Global Average Pooling (GAP):** Reduces feature maps.
 - **Dropout:** 50% dropout to mitigate overfitting.
 - **Dense Layer:** Outputs probabilities for 5 classes using soft max activation.
4. **Training:**
 - **Phase 1:**
 - EfficientNetB0 layers are frozen, and only the custom head is trained.
 - Optimizer: Adam with a learning rate of 0.001.
 - **Phase 2:**
 - Unfreeze the entire EfficientNetB0 backbone and fine-tune with a lower learning rate (0.0001).
5. **Callbacks:**
 - **Epoch Time Logger:**
 - Logs time taken, training loss, validation loss, and accuracies per epoch.
 - **Tqdm Callback:**
 - Provides a progress bar for better visualization.
6. **Evaluation:**
 - **Predictions:**
 - The model generates predictions on the validation dataset.
 - **Kappa Score:**
 - Quadratic Weighted Kappa Score measures agreement between predicted and true labels.



4. Implementation:

The provided solution is aimed at training a deep learning model to classify diabetic retinopathy (DR) severity using EfficientNetB0, a pre-trained convolutional neural network, as a base. The model is fine-tuned in two stages:

- **Initial Training:** The EfficientNet base model is frozen, and only the top layers are trained.
- **Fine-tuning:** The entire EfficientNet model is unfrozen, and all layers are fine-tuned with a smaller learning rate.

4.1 Proposed methodology:

The solution follows a systematic approach to solve the problem of diabetic retinopathy severity classification:

1. **Data Preprocessing:**
 - a. Image paths are loaded from train.csv, and the file names are appended with .png.
 - b. Data is split into **training** and **validation** sets using train_test_split with stratified sampling to ensure class balance.
2. **Data Augmentation:**
 - a. Training data is augmented using ImageDataGenerator with transformations like rotation, zoom, shifts, flips, and shear.
 - b. Validation data undergoes rescaling only.
3. **Model Design:**
 - a. EfficientNetB0 is used as a base model with pre-trained ImageNet weights.
 - b. The top layers include:
 - i. Global Average Pooling layer for feature extraction.
 - ii. Dropout layer to reduce overfitting.
 - iii. Dense layer with softmax activation to predict the 5-class output.
4. **Two-Phase Training:**
 - a. **Phase 1:** Base model layers are frozen to train only the custom top layers at a learning rate of 0.001.
 - b. **Phase 2 (Fine-tuning):** All layers are unfrozen, and the entire model is trained with a smaller learning rate (0.0001).
5. **Monitoring and Evaluation:**
 - a. Custom EpochTimeLogger callback is implemented to measure training time per epoch and log performance metrics.
 - b. TqdmCallback is integrated for enhanced progress visualization.

- c. The model is evaluated using the **Quadratic Weighted Kappa Score**, which is a standard metric for ordinal classification like diabetic retinopathy grading.

4.2 Algorithm Used for Implementation:

1. EfficientNetB0 Architecture:

- EfficientNetB0 is used as the base model, pre-trained on the **ImageNet** dataset.
- The model balances **depth**, **width**, and **resolution** using a **compound scaling** method for efficiency and accuracy.

2. Transfer Learning:

- Freeze the pre-trained **EfficientNetB0** base layers to act as a feature extractor.
- Add a **Global Average Pooling layer**, a **Dropout layer**, and a **Dense softmax classifier** with 5 output classes (diabetic retinopathy severity levels).
- Train only the top layers initially with a learning rate of **0.001**.

3. Fine-Tuning:

- Unfreeze the **base model layers** after transfer learning.
- Retrain the entire model with a smaller learning rate (**0.0001**) to adapt the deeper features to the specific dataset.

4. Quadratic Weighted Kappa Score:

- Evaluate model performance using the **QWK score**, which measures agreement between predictions and ground truth.
- QWK considers the **ordinal nature** of the classes and penalizes predictions proportionally based on the distance from the correct class.

5. Data Augmentation:

- Apply augmentations such as rotation, zoom, shifts, shear, and horizontal flips to improve generalization.

6. Training:

- Use **ImageDataGenerator** to preprocess and load images in batches for training and validation.
- Train the model for two phases: **transfer learning (frozen base)** and **fine-tuning (unfrozen base)**.

7. Model Saving and Evaluation:

- Save the trained model.
- Predict on the validation set and compute the **QWK score** for final performance evaluation.

4.3 Tools and Technologies Used:

- **Programming Language:** Python
- **Deep Learning Framework:** TensorFlow/Keras
- **EfficientNetB0:** Pre-trained deep learning model for feature extraction.
- **Pandas:** For handling and preprocessing tabular data.
- **Scikit-learn:** Used for train-test splitting and calculating the Kappa score.
- **ImageDataGenerator:** For data augmentation and loading images from directories.
- **Callbacks:**
 - Custom EpochTimeLogger for epoch timing and metric logging.
 - TqdmCallback for improved progress visualization.
- **NumPy:** For numerical operations and predictions post-processing.
- **Operating System Module (os):** For handling file paths.

4.4 Testing

1. Data Splitting:

a. Stratified Sampling:

- Used stratified sampling to maintain class distribution across training and validation sets, preventing class imbalance issues.

2. Image Data Pipeline:

a. Augmentation Verification:

- Ensured transformations like **rotation, zoom, shifts, shear, and flips** diversify images without distorting visual content.

b. Data Loading:

- Verified `flow_from_dataframe()` loads images correctly with consistent target size (**224x224**) and batch size (**32**).

c. Validation Pipeline:

- Ensured validation images were correctly rescaled to match training input format.

3. Model Training:

a. Transfer Learning:

- i. Stabilized training by freezing **EfficientNetB0** base layers, allowing only the custom layers to train initially.
- b. **Fine-Tuning:**
 - i. Gradually unfroze base model layers and reduced the learning rate to **0.0001** for precise weight adjustments.
 - ii. Monitored validation loss to ensure fine-tuning improved generalization without overfitting.

4. Callbacks and Logs:

- a. **EpochTimeLogger:**
 - i. Verified logs for epoch **duration**, training/validation **loss**, and **accuracy** at the end of each epoch.
- b. **Progress Tracking:**
 - i. Used TqdmCallback for real-time progress visualization during training.

5. Performance Evaluation:

- a. **Model Predictions:**
 - i. Predicted validation set outputs and converted them into class labels using `argmax()`.
- b. **Quadratic Weighted Kappa:**
 - i. Calculated **QWK score** to measure performance, considering the **ordinal nature** of classes and penalizing distant misclassifications.
- c. **Final Validation:**
 - i. Confirmed model generalizes well by comparing predictions to ground truth labels and reporting the **final QWK score**.

5. Results and Discussion

This section presents the results obtained from the diabetic retinopathy classification model using the EfficientNetB0 architecture, along with a detailed discussion of the findings. The results include metrics such as **training accuracy**, **validation accuracy**, **loss curves**, and the **Quadratic Weighted Kappa Score**, which is the primary evaluation metric.

5.1 Model Performance Metrics

The model was trained in two phases:

1. **Feature Extraction Phase** (Freezing the EfficientNetB0 backbone).
2. **Fine-Tuning Phase** (Unfreezing the backbone for end-to-end training).

The performance metrics observed during these phases are summarized in the table below:

Phase	Accuracy	Loss	Kappa Score
Training Phase	88.3%	0.2678	0.87
Validation Phase	89.5%	0.2425	0.89
Fine-Tuning (Validation)	92.1%	0.1987	0.91

- **Training Accuracy** improved steadily from **88.3%** during feature extraction to **92.1%** after fine-tuning.
- **Validation Accuracy** reached **89.5%** during initial training and further improved to **92.1%** after fine-tuning.
- The **Quadratic Weighted Kappa Score** also showed strong results, achieving **0.91** in the fine-tuning phase, indicating excellent agreement with ground truth labels.

5.2 Accuracy and Loss Curves

The accuracy and loss curves during training and validation phases are shown below:

1. **Training and Validation Accuracy Curve:**
 - a. The accuracy consistently increases over epochs, demonstrating that the model is learning effectively.
 - b. A slight improvement is observed after unfreezing the backbone during fine-tuning.

2. **Training and Validation Loss Curve:**

- a. The loss steadily decreases across epochs, showing that the model's predictions become more accurate as training progresses.
- b. Validation loss remains close to training loss, suggesting no significant overfitting.

5.3 Discussion

The following observations can be made based on the results:

1. **Effectiveness of EfficientNetB0:**

The EfficientNetB0 backbone effectively extracts features from retinal fundus images, which contributes significantly to the model's accuracy. The pre-trained weights on ImageNet provided a strong initialization for transfer learning.

2. **Impact of Data Augmentation:**

Data augmentation techniques, including rotation, zooming, and horizontal flips, increased the robustness of the model. By simulating variations in retinal images, the model generalizes well to unseen validation data.

3. **Fine-Tuning Benefits:**

Unfreezing the EfficientNetB0 backbone during fine-tuning improved the model's accuracy from **89.5%** to **92.1%**, and the Kappa Score improved to **0.91**. Fine-tuning allowed the entire network to adapt to the specific features of diabetic retinopathy.

4. **Evaluation using Kappa Score:**

The **Quadratic Weighted Kappa Score** of **0.91** highlights the model's ability to accurately classify the severity of diabetic retinopathy. This metric accounts for the ordinal nature of the problem, where misclassifying a severe case as mild is more significant than a slight misclassification.

5. **Training Efficiency:**

The custom **Epoch Time Logger** callback provided insights into the model's efficiency, with each epoch completing in approximately **50-60 seconds** on an appropriate GPU setup.

6. Conclusion Future Work

6.1 Conclusion

The diabetic retinopathy classification model presented in this work demonstrates the effectiveness of leveraging **EfficientNetB0**, a state-of-the-art convolutional neural network, combined with a custom classification head for severity level prediction. The model achieved strong performance with a **Validation Accuracy of 89.5%** and a **Quadratic Weighted Kappa Score of 0.89**. By adopting a two-phase training strategy, where the backbone is initially frozen during feature extraction and later fine-tuned, the model was able to efficiently generalize while avoiding overfitting. Data augmentation techniques further enhanced the model's robustness by introducing variations in the training data.

The results validate that deep learning-based approaches can significantly aid in automating diabetic retinopathy diagnosis, offering a scalable and reliable solution to assist healthcare professionals in early detection and monitoring of the disease.

6.2 Future Work

While the current model achieves promising results, there are several avenues for further optimization and exploration to improve its performance and scalability:

1. **Hyperparameter Optimization:**
 - a. Implement advanced hyperparameter tuning techniques such as **Bayesian Optimization** or **Grid Search** to identify the optimal learning rate, batch size, dropout rate, and optimizer configurations.
 - b. Use tools like **Keras Tuner** or **Optuna** for automated tuning.
2. **Ensemble Learning:**
 - a. Combine multiple deep learning architectures (e.g., EfficientNet, ResNet, DenseNet) to create an ensemble model that improves robustness and reduces prediction errors.
 - b. Use techniques like weighted averaging or majority voting for ensemble predictions.
3. **Incorporating Attention Mechanisms:**
 - a. Integrate attention-based methods such as **SE (Squeeze-and-Excitation) blocks** or **Vision Transformers (ViT)** to focus on the most relevant regions of the retinal images. This can enhance the model's ability to identify subtle features indicative of diabetic retinopathy.
4. **Multi-Scale Feature Extraction:**
 - a. Explore architectures that process images at multiple resolutions or scales to capture both global and local features more effectively.
5. **Class Imbalance Mitigation:**
 - a. Address class imbalance in the dataset using techniques like **class weighting**, **oversampling minority classes**, or **SMOTE (Synthetic Minority Oversampling Technique)** to further improve performance on underrepresented classes.
6. **Integration of Explainable AI (XAI):**

- a. Use explainable AI techniques such as **Grad-CAM** or **LIME** to visualize the regions of images that contribute most to predictions. This will improve the interpretability of the model, which is critical in medical applications.

7. **Deployment and Scalability:**

- a. Optimize the model for deployment on edge devices or cloud platforms by using tools like **TensorFlow Lite** or **ONNX** for reduced latency and computation costs.
- b. Develop a web-based or mobile-based application to enable real-time diagnosis in clinical settings.

8. **Dataset Expansion and Generalization:**

- a. Train the model on larger, more diverse datasets to ensure better generalization across populations and imaging devices.
- b. Incorporate domain adaptation techniques to handle variations in image quality and lighting.

7. References:

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